Kathmandu University School of Management

Balkumari, Lalitpur, Nepal



Report on: "**Analysis of E-commerce data"**

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Evidence Based Data Analytics (MAS411)

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Submitted to:

Uttam Karki

Faculty of Computer Science

Submitted by:

Group: 9

Risita Sedain (217086)

Samriddhi Acharya (217122)

Shreya Subedi (207106)

Simran Regmi (217084)

Sushovan Paudel (217)

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EXECUTIVE SUMMARY

This report analyzes customer behavior patterns in e-commerce transactions using data extracted from Kaggle. The study focuses on key consumer metrics such as age-based spending, membership impact on purchase volume, discount-driven purchases, geographic spending variations, and the relationship between item count and total spend. Using Power BI for data cleaning and visualization, we identified meaningful trends that help businesses optimize marketing, pricing, and customer engagement strategies. The findings reveal that younger consumers (25-34) have higher average spending, and Gold-tier members purchase more items on average than Silver and Bronze members. Discounted records have lower average spending, but the effect of discounts cannot be established. Spending varies by city, and item count is positively associated with total spend. These patterns can inform further study of marketing, membership and pricing strategies.

CHAPTER ONE: INTRODUCTION AND PROBLEM STATEMENT

1.1 Introduction

E-commerce has revolutionized consumer purchasing behavior by providing a convenient, efficient, and data-driven shopping experience. As businesses shift towards online platforms, understanding customer behavior patterns has become critical for enhancing customer engagement, optimizing pricing strategies, and personalizing marketing efforts. With vast amounts of transaction data available, businesses can analyze consumer interactions to identify trends that drive revenue growth and improve user experience.

This study focuses on customer behavior patterns in e-commerce transactions, analyzing key metrics such as age-based spending trends, membership influence on purchase volume, discount effects on spending, city-wise purchasing behavior, and item count versus total spend. Using data extracted from Kaggle, we employ Power BI for data cleaning and visualization, enabling a comprehensive analysis of purchasing behavior. The results provide data-driven insights for businesses to develop strategic marketing, pricing, and retention approaches that align with consumer tendencies and preferences.

1.1 Key Research Problems

- Understanding customer purchasing behavior is crucial for optimizing marketing strategies, inventory management, and personalized recommendations.
- This study explores customer transaction data to uncover insights into spending habits, peak sales periods, and product preferences.
- By analyzing historical sales data, we aim to improve decision-making regarding pricing, promotions, and inventory stocking.

1.2 Key Goals

1. Identify key purchasing trends and high-value buyers.
2. Determine peak sales periods and daily sales activity.
3. Analyze the impact of discounts on customer spending behavior.
4. Examine regional sales distribution and customer segmentation.

5.Derive actionable insights to enhance business strategy.

CHAPTER TWO: DATA SOURCES

2.1 Data Collection

The analysis is based on structured data covering:

- **Customer Demographics:** Age, Gender, City
- **Spending Behavior:** Total Spend, Items Purchased, Discount Applied
- **Customer Satisfaction:** Satisfaction Level, Average Rating
- **Engagement Metrics:** Days Since Last Purchase, Membership Type

CHAPTER THREE: HYPOTHESES AND ASSUMPTIONS

Hypothesis 1: Age and Spending Behavior

Hypothesis: Younger individuals spend more on e-commerce purchases than older individuals.

Rationale: Younger consumers may be more engaged in e-commerce because of digital convenience. The data shows higher average spending among customers aged 25-34 than among those aged 35-44. It does not establish the reasons for this difference or measure disposable income.

Assumptions:

- Younger individuals are more tech-savvy and engaged in digital shopping.
- Older individuals may prefer offline retail or have more conservative spending habits.
- The dataset accurately represents purchasing behaviors across different age groups.

Hypothesis 2: Membership Type and Purchase Volume

Hypothesis: Customers with higher-tier memberships (Gold) purchase more items than those in lower-tier memberships (Silver, Bronze).

Rationale: Membership programs may incentivize spending through loyalty rewards, discounts and exclusive deals. Gold members are expected to purchase more. The dataset shows that Gold members purchase more items on average than Silver and Bronze members, but it does not measure purchase frequency or the reasons for the difference.

Assumptions:

- Membership tiers accurately reflect customer engagement.
- Gold members have the financial capacity for higher spending.
- External promotions do not heavily influence Silver/Bronze members' purchasing decisions.

Hypothesis 3: Discounts Lead to Increased Spending
Hypothesis: Customers spend more when discounts are applied.

Rationale: Discounts may encourage purchases by making products more attractive. In this dataset, discounted records have lower average total spend than non-discounted records. This comparison does not establish the causal effect of discounts or customers' reasons for purchasing.

Assumptions:

- Discounted items are desirable to customers.
- Customers perceive discounts as an opportunity for savings, not just cost reduction.
- Discounted purchases do not lead to stockpiling behavior reducing future spending.

Hypothesis 4: City-Based Spending Patterns
Hypothesis: Customers from metropolitan cities spend more than those from smaller cities.

Rationale: Urban areas may differ in income and shopping options. The dataset shows variation in spending across cities, but does not classify cities as metropolitan or smaller. The reasons for these differences cannot be established from the available data.

Assumptions:

- Urban customers have greater purchasing power.
- Smaller cities lack access to competitive offline retail options.
- Data is not biased toward any particular region.

Hypothesis 5: Number of Items Purchased vs. Total Spend
Hypothesis: Customers who purchase more items tend to have a higher total spend.

Rationale: Conventional retail logic suggests that purchasing more items results in higher total spending. Python analysis of the customer records shows a strong positive association between item count and total spend. The aggregate chart varies with the number of records at each item count and does not establish causation or proportionality.

Assumptions:

- Higher item counts correlate with a greater total basket value.
- Discounts do not significantly affect per-unit pricing.
- Customers do not prioritize lower-cost items when purchasing in bulk.

CHAPTER FOUR: EXPLORATORY DATA ANALYSIS (EDA)

4.1 Data Cleaning

- The dataset was extracted from Kaggle.
- Data was cleaned using Power BI and Python during the transformation process.
- Handled missing values and standardized data formats for consistency.

Step 1: Import Necessary Libraries

First, we imported the necessary libraries for data analysis and visualization.

Step 2: Load the Dataset

Assuming the dataset is in a CSV file, we can load it using pandas.

Step 3: Data Cleaning

The report mentions that data cleaning was done using Power BI. In Python, we can handle missing values and standardize data formats.

Step 4: Exploratory Data Analysis (EDA)

We performed EDA to understand the data better.

4.2 Data Visualization

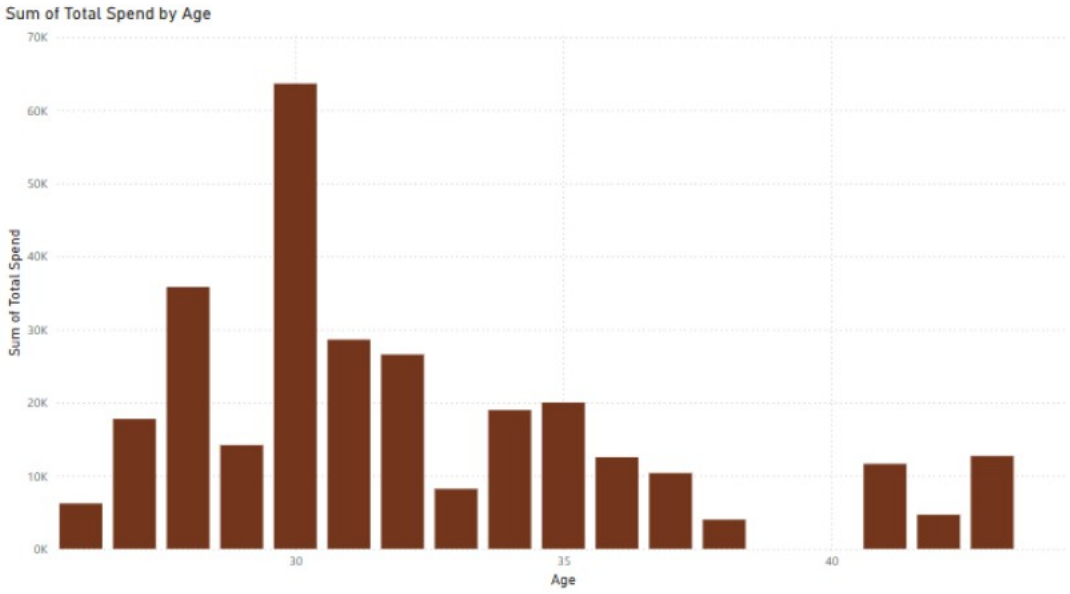
- Power BI was used for data visualization and dashboard creation.
- **Line Charts:** Used to visualize trends in customer purchases over time.
- **Ribbon Chart:** Showed the ranking of top-spending customer segments.
- **Bar Graphs:** Compared spending across different membership tiers and customer categories.
- **Column Stacked Chart:** Analyzed the impact of discounts on purchase behavior.

CHAPTER FIVE: DATA VISUALIZATION IN POWER BI AND ANALYSIS

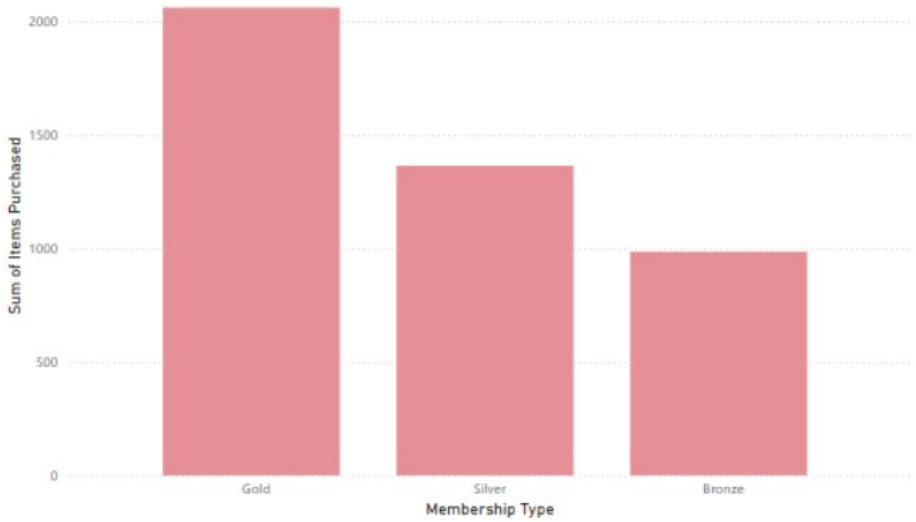
5.1 Dashboards Created

- **Line Charts:** Days Since Last Purchase vs. Customer Retention
- **Ribbon Charts:** Ranking of High-Spending Customer Segments
- **Bar Graphs:** Membership Type vs. Total Spend
- **Column Stacked Charts:** Discount Applied vs. Items Purchased

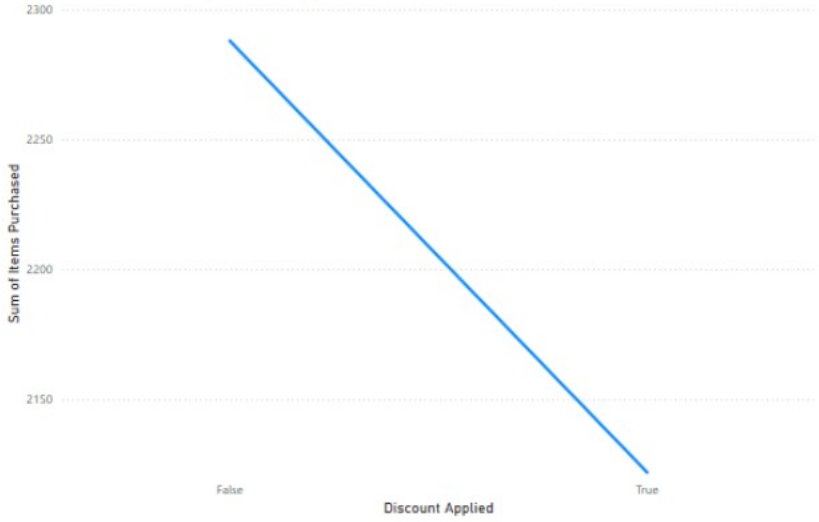
5.2 Data Visualization

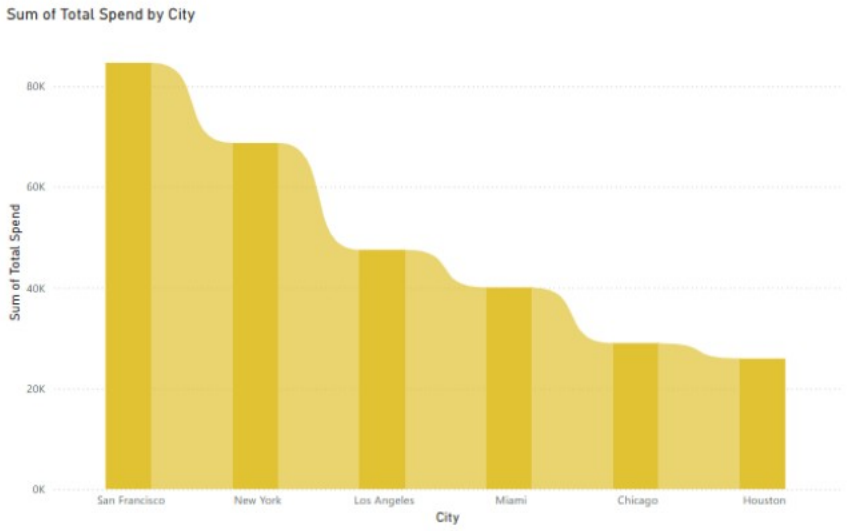


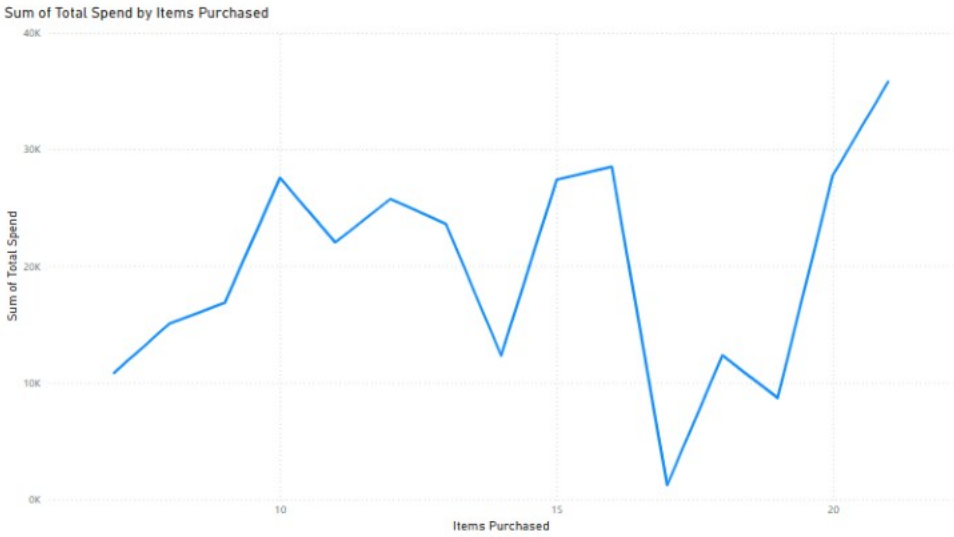
Sum of Items Purchased by Membership Type



Sum of Items Purchased by Discount Applied







5.3 Analysis of Hypotheses

Hypothesis 1: Age and Spending Behavior

Supported: Yes (descriptively)
Explanation: The bar chart shows total spend by individual age. Python analysis shows that customers aged 25-34 have higher average spend than those aged 35-44. This association does not establish that digital engagement or marketing caused the difference.

Hypothesis 2: Membership Type and Purchase Volume

Supported: Yes (descriptively)
Explanation: The column chart shows total items purchased by membership type. Python analysis shows that Gold members average 17.62 items, compared with 11.66 for Silver and 8.49 for Bronze. The data does not establish purchase frequency or the effect of promotions.

Hypothesis 3: Discounts Lead to Increased Spending

Supported: No (descriptively)
Explanation: The chart shows total items purchased by discount status, not a box plot of spending. Average spend is 787.27 for discounted records and 903.49 for non-discounted records. This comparison does not establish a causal effect or statistical significance.

Hypothesis 4: City-Based Spending Patterns

Supported: Not established
Explanation: The chart shows total spend by city, with San Francisco and New York highest. The dataset does not classify cities as metropolitan or smaller, so the stated hypothesis cannot be assessed directly.

City and membership type are closely linked in these records, and the data does not establish the reasons for city differences.

Hypothesis 5: Number of Items Purchased vs. Total Spend

Supported: Yes (descriptively)
Explanation: Python analysis finds a strong positive association between items purchased and total spend (Pearson r = 0.972). The original line chart shows the sum of spend for each item count, so its dips also reflect different group sizes. This does not prove causation or proportionality.

CHAPTER SIX: FINDINGS & DISCUSSIONS

6.1 Key Findings

Customer spending behavior varies across demographic and engagement factors. Younger consumers aged 25-34 have higher average spend than those aged 35-44. Gold members have the highest average number of items purchased. Discounted records have lower average spend, but the effect of discounts cannot be determined from this comparison. Spending varies by city, although the dataset does not classify city size. Items purchased and total spend show a strong positive association; the underlying causes are not established.

6.2 Managerial Insights

1. **Enhance Membership Perks:** Gold members have higher average spending; consider evaluating incentives for lower-tier customers.

2. **Targeted Promotions by Age Group:** Design promotions tailored to middle-aged customers.

3. **Optimize Discounts:** Test discount strategies and measure their effects on purchases and profit margins.

4. **Improve Customer Experience:** Explore the relationship between satisfaction and ratings before drawing conclusions.

5. **Loyalty Programs:** Explore ways to reduce purchase gaps using additional customer purchase-history data.

CHAPTER SEVEN: CONCLUSION & LIMITATIONS

7.1 Summary of Findings

This study provides insights into customer behavior in e-commerce. Younger customers and Gold members have higher average spending in this dataset. Discounted records do not have higher average spending, but the causal effect of discounts is unknown. Spending varies across cities, and the data does not establish differences by city size. Items purchased and total spend are positively associated. These findings can guide further investigation of promotions, membership programs and customer engagement.

- Customer behavior varies based on membership type, age, discounts, and location.
- Power BI visualizations help identify meaningful relationships that aid business decisions.

7.2 Limitations

- External economic factors may influence spending habits beyond what is captured in the dataset.
- The dataset does not provide insights into brand preferences or product-specific behavior.

7.3 Future Work

- Implement AI-driven recommendation systems to enhance personalization.

- Conduct A/B testing to measure the effectiveness of different promotions.
- Incorporate sentiment analysis from customer feedback to enhance insights.

CHAPTER EIGHT: REFERENCES

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Python reanalysis data:
https://www.kaggle.com/datasets/uom190346a/e-commerce-customer-behavior-dataset

Annexure

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load the dataset
df = pd.read_csv('ecommerce_data.csv')

# Display the first few rows of the dataset
df.head()
```

```
# Check for missing values
df.isnull().sum()

# Handle missing values (e.g., fill with mean or drop)
df.fillna(df.mean(), inplace=True)

# Standardize data formats (e.g., convert date columns to datetime)
df['Purchase_Date'] = pd.to_datetime(df['Purchase_Date'])
```

```
# Summary statistics
df.describe()

# Distribution of Age
sns.histplot(df['Age'], bins=20, kde=True)
plt.title('Age Distribution')
plt.show()

# Total Spend by Age Group
sns.barplot(x='Age_Group', y='Total_Spend', data=df)
plt.title('Total Spend by Age Group')
plt.show()
```